

Pandas & Matplotlib Practice Questions

Pandas Practice — 20 Questions

- Q1.** Load the dataset into a DataFrame and display the first 5 rows.
- Q2.** Display the last 10 rows of the DataFrame.
- Q3.** Display the first 8 rows and last 8 rows together for inspection.
- Q4.** Find the number of rows and columns in the dataset.
- Q5.** Display all column names and identify which columns contain numerical data.
- Q6.** Display the data types of all columns.
- Q7.** Find the number of unique values in each categorical column.
- Q8.** Display the unique values present in the Gender column.
- Q9.** Find the number of customers belonging to each Product Category.
- Q10.** Find the average Age of all customers.
- Q11.** Find the average Annual Income for each Gender using groupby().
- Q12.** Find the maximum Spending Score for each Product Category using groupby().
- Q13.** Find the minimum and maximum Annual Income for each Gender.
- Q14.** Find the average Age and average Spending Score for each Gender using a single groupby operation.
- Q15.** Find the number of customers in each City using groupby().
- Q16.** Find the average Annual Income for each Product Category and sort the result from highest to lowest.
- Q17.** Find the average Spending Score for each Gender and Product Category using groupby() with two grouping columns.
- Q18.** Find the Product Category having the highest average Spending Score.
- Q19.** Find the Gender having the highest average Annual Income.
- Q20.** Create a grouped summary showing count, mean, minimum, and maximum of Annual Income for each Product Category.

Matplotlib Practice — 20 Questions

- Q1.** Create a line graph showing the Age values of customers in their row order.
- Q2.** Create a line graph showing Spending Score values and add a suitable title and axis labels.
- Q3.** Create a histogram to show the distribution of Age.
- Q4.** Create a histogram for Annual Income and experiment with different numbers of bins.
- Q5.** Create a bar chart showing the number of customers in each Gender.
- Q6.** Create a bar chart showing the number of customers in each Product Category.
- Q7.** Create a bar chart showing the number of customers in each City and rotate the x-axis labels.
- Q8.** Using `groupby()`, calculate the average Annual Income by Gender and visualize the result with a bar chart.
- Q9.** Using `groupby()`, calculate the average Spending Score by Product Category and visualize it with a bar chart.
- Q10.** Using `groupby()`, calculate the average Age for each Gender and customize the graph title, labels, figure size, and grid.
- Q11.** Create a bar chart of the average Annual Income by Product Category, sorted from highest to lowest.
- Q12.** Create a pie chart showing the proportion of customers in each Gender.
- Q13.** Create a pie chart showing the proportion of customers in each Product Category.
- Q14.** Create a line graph comparing the average Spending Score across Product Categories after calculating the averages with `groupby()`.
- Q15.** Create a bar chart comparing average Age and average Spending Score for each Gender.
- Q16.** Create a grouped bar chart comparing Male and Female customer counts across Product Categories.
- Q17.** Create a histogram of Spending Score and customize its figure size, number of bins, transparency, and grid.
- Q18.** Create a bar chart of City-wise customer counts and customize the title, axis labels, tick rotation, figure size, and grid.
- Q19.** Create a graph for the average Annual Income by Gender and add a suitable legend, annotations, and customized tick labels.
- Q20.** Create a final EDA dashboard-style figure containing four different visualizations from the dataset using subplots. Do not use boxplots or scatter plots.